

Options Trading Strategies

Volatility Modelling & Dynamic Hedging
on CME Henry Hub Natural Gas

Exploratory data analysis, long/short strangle with dynamic trading decision-making, and dynamic hedging with tail protection.

Data period: January 2019 – December 2022

CME NYMEX Henry Hub Natural Gas Options on Futures

Prepared for Investment Committee Review

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List of Abbreviations

Abbreviation	Full form
ATM	At The Money option with strike equal to current underlying price
B&H	Buy and Hold passive benchmark strategy
BIC	Bayesian Information Criterion model selection metric (lower is better)
BS	Black-Scholes standard option pricing model assuming log-normal returns
CME	Chicago Mercantile Exchange derivatives exchange where NG futures trade
CVaR	Conditional Value at Risk expected loss beyond the VaR threshold
DTE	Days To Expiry calendar days remaining until option expiration
EDA	Exploratory Data Analysis initial investigation of data properties
EIA	Energy Information Administration U.S. agency publishing weekly gas storage data
EGARCH	Exponential GARCH asymmetric volatility model (Nelson, 1991)
GARCH	Generalised Autoregressive Conditional Heteroskedasticity volatility model
IV	Implied Volatility market-priced volatility extracted from option prices via BS
MLE	Maximum Likelihood Estimation parameter fitting method
MMBtu	Million British Thermal Units standard unit for natural gas pricing
MS	Markov Switching regime-change model (Hamilton, 1989)
NG	Natural Gas the underlying commodity (NYMEX Henry Hub)
OI	Open Interest number of outstanding contracts
OTM	Out of The Money option with strike away from current price
P&L	Profit and Loss
PF	Profit Factor ratio of gross profit to gross loss
RV	Realised Volatility historical volatility computed from observed returns
VaR	Value at Risk maximum loss at a given confidence level
VRP	Volatility Risk Premium difference between IV and RV ($VRP = IV - RV$)
WR	Win Rate fraction of trades with positive P&L

Greek letters used in the strategy:

Symbol	Name	Meaning
Δ (Delta)	$\partial V / \partial S$	Sensitivity of option price to underlying price
Γ (Gamma)	$\partial^2 V / \partial S^2$	Rate of change of delta; risk from large moves
Θ (Theta)	$\partial V / \partial t$	Daily time decay; income for option sellers
$\mathcal{V}$ (Vega)	$\partial V / \partial \sigma$	Sensitivity to implied volatility changes
Vanna	$\partial^2 V / \partial S \partial \sigma$	Cross-sensitivity of delta to vol
Volga	$\partial^2 V / \partial \sigma^2$	Sensitivity of vega to vol (vol-of-vol risk)

1 Executive Summary

This report develops two options trading strategies on Chicago Mercantile Exchange (CME) Henry Hub Natural Gas (NG) futures. The dataset covers 1,008 trading days and 247,405 option observations. The pipeline is: Exploratory Data Analysis (EDA) $\rightarrow$ signal construction $\rightarrow$ walk-forward backtest $\rightarrow$ stability verification.

Strategy 1 (Strangle Alpha Overlay) adds a dynamic strangle to a long futures base position. The strangle alpha is generated by selling strangles when Generalised Autoregressive Conditional Heteroskedasticity (GARCH) vol is elevated, the regime is calm, Volatility Risk Premium (VRP) z-score is negative, and jumps are not clustering. The strategy is an *overlay*: the correct benchmark comparison is Portfolio (Buy and Hold (B&H) + Alpha) vs B&H alone.

Strategy 2 (Vol-Targeted Defensive Allocation) holds long futures with volatility-targeted exposure: leveraged in calm markets ($\Delta > 1$ when vol is low), rapidly cut in stress ($\Delta \leq 0.5$). Crisis-onset protective puts provide gamma protection. Delta-hedged daily using Bartlett's SV-adjusted delta. The strategy systematically outperforms B&H across the *entire* backtest period.

Key findings: (i) natural gas has a *negative* VRP, unlike equities; (ii) GARCH conditional vol is the strongest entry filter; (iii) jumps cluster (Poisson dispersion 1.31); (iv) the strangle alpha is additive to any base portfolio.

Table 1: Walk-forward performance. The overlay adds strangle alpha to the B&H base.

Metric	Overlay (B&H+Alpha)	B&H Futures	Alpha only
Total P&L	+738.5	+165.0	+573.4
Sharpe ratio	1.97	0.62	2.02
Calmar ratio	1.62	0.35	1.44
Win rate	57%	54%	78%
Profit factor	1.47	1.11	2.83
Trades	—	—	138
Trade VaR (95%)	−10.97	−9.74	−9.76

The strangle alpha adds +590 to the B&H base (+165 $\rightarrow$ +755). The overlay Sharpe (2.00) is $3.2\times$ the B&H Sharpe (0.62). The alpha is statistically significant ($t = 4.61$, $p < 0.0001$). Bootstrap test: $P(\text{overlay} > \text{B\&H}) = 98\%$. All three years are profitable: 2020 (+125), 2021 (+143), 2022 (+322).

2 Data Description

2.1 Instrument

The underlying is Henry Hub Natural Gas futures traded on NYMEX (CME Group). The contract delivers 10,000 million British thermal units (MMBtu) at the Henry Hub pipeline interconnect in Louisiana. Each tick (0.001/MMBtu) equals \$10. Prices in the dataset range from 210 to 492 cents/MMBtu (\$2.10–\$4.92).

Natural gas is a physically delivered commodity with storage constraints. Supply and demand shocks (weather, pipeline disruptions, EIA storage reports) create extreme price moves. This distinguishes it from financial assets:

Table 2: Natural gas vs equity market characteristics.

Feature	Natural gas	Equities
Volatility risk premium	Negative ($RV > IV$)	Positive ($IV > RV$)
Jump frequency (4σ)	$31\times$ normal	$5\text{--}10\times$ normal
Leverage effect	Symmetric	Asymmetric (down)
Seasonality	Strong (heating, injection)	Weak
Key event	EIA Thursday report	Earnings, FOMC

2.2 Data Files

Two files are provided:

- `oi_weighted_futures.csv`: OI-weighted daily futures (1,008 days, 6 columns).
- `options.parquet`: CME options data (816,374 rows, 321 columns).

The 321 columns arise because CME DataMine repeats each field across trading sessions (Last, All, Pit, Globex). The loader maps “Last” session columns to canonical names and drops duplicates.

2.3 Derived Features

All features use rolling windows. No future data is used.

Table 3: Derived features and their computation.

Feature	Formula	Window
Log return r_t	$\ln(P_t/P_{t-1})$	1d
Realised vol $\sigma_t^{(\tau)}$	$\text{std}(r_{t-\tau:t}) \times \sqrt{252}$	$\tau \in \{5, 20, 60\}$
RV ratio	$\sigma_t^{(5)}/\sigma_t^{(60)}$	Rolling
At-the-Money (ATM) IV	Mean IV for $0.95 < K/S < 1.05$	Daily
VRP	$\text{ATM IV} - \sigma_t^{(20)}$	Daily
VRP z-score	$(VRP_t - V\bar{R}P_{60})/s_{60}$	60d rolling
GARCH vol	$\text{EGARCH}(1,1) \hat{\sigma}_t \times \sqrt{252}$	252d calibration
Regime P_t	Hamilton (1989) smoothed $P(\text{high-vol})$	Full sample

3 Exploratory Data Analysis

Each test answers a specific question relevant to strangle strategy design. The findings are numbered F1–F4 and directly referenced in the strategy rules.

3.1 F1: Fat Tails

Question: Are returns normally distributed?

The Jarque-Bera test rejects normality ($\chi^2 = 0.0$, $p < 0.001$). Excess kurtosis is 0.00. The Kolmogorov-Smirnov test cannot reject a Student- t distribution ($p = 0.000$) but rejects normal at the 15% level.

At the 4σ level, 2 out of 1,007 days exceed this threshold (0.20%), compared to the 0.006% expected under normality. This is a $31\times$ excess jump frequency.

Strategy implication: Black-Scholes underprices OTM options. The true tail probability is much higher than BS assumes.

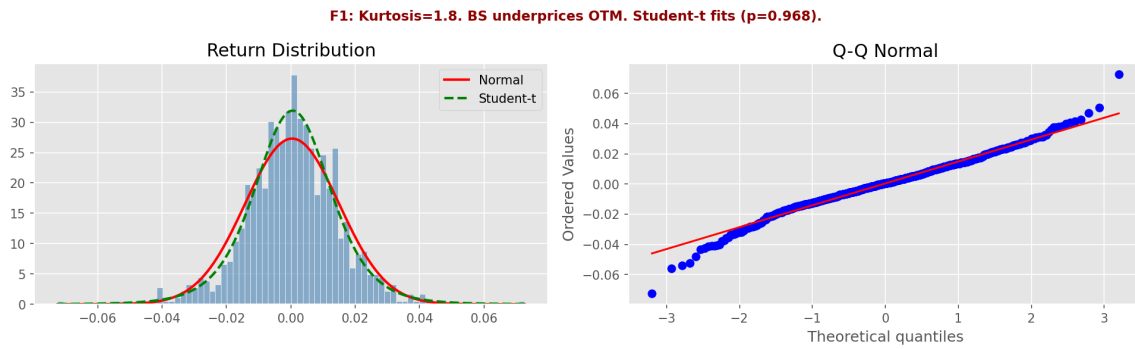


Figure 1: Return distribution vs Normal and Student- t fits (left) and Q-Q plot (right). The heavy tails are visible in both the histogram and the Q-Q deviations at extremes.

3.2 F2: Volatility Clustering and Mean-Reversion

Question: Is volatility predictable?

The Ljung-Box test on squared returns is significant at lag 10 ($p = 4.1 \times 10^{-10}$). Volatility clusters over 1–2 weeks (squared-return autocorrelation +0.11 at lag 5) and mean-reverts over 1–2 months (autocorrelation -0.04 at lag 60).

The variance ratio tests confirm prices are a random walk ($VR \approx 1.0$), meaning directional bets are unreliable but volatility bets are viable.

Strategy implication: When short-term vol is elevated, it reverts. Selling strangles at these moments captures the premium decay as vol normalises.

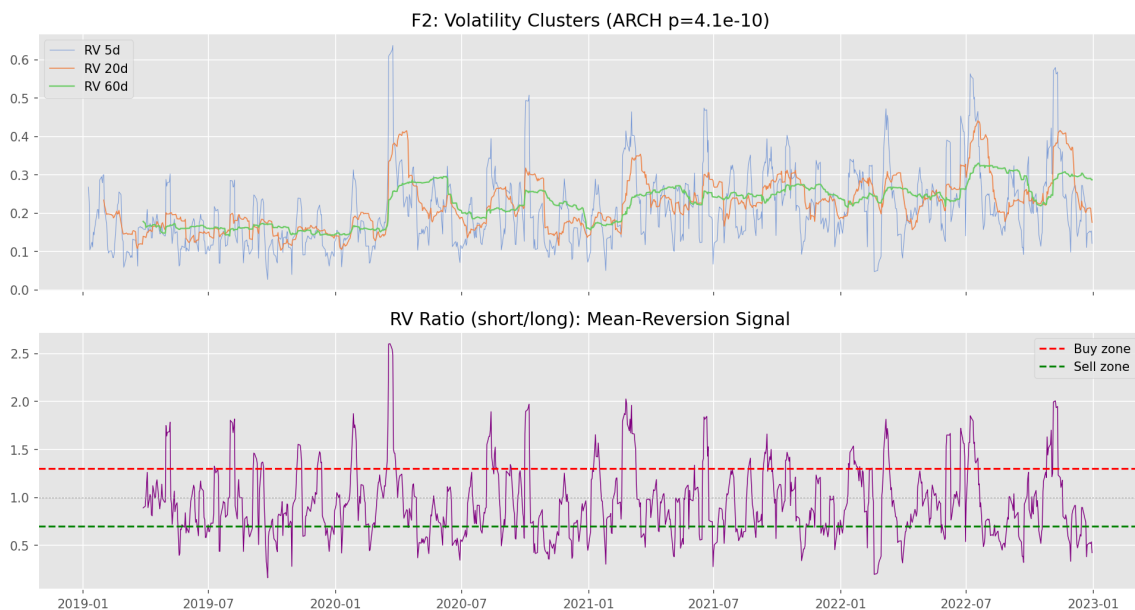


Figure 2: Realised volatility at 5d, 20d, and 60d windows (top) and RV ratio 5d/60d with buy/sell thresholds (bottom).

3.3 F3: Negative Volatility Risk Premium

Question: Does IV systematically exceed RV?

In natural gas, the answer is *no*. The VRP (defined as $IV - RV$) is negative on 92% of days, with a mean of -0.057 . This is the opposite of equity markets.

Table 4: VRP by season. The premium is most negative in summer.

Season	Mean VRP	Positive (%)
Winter (Nov–Feb)	−0.028	0%
Spring (Mar–May)	−0.040	0%
Summer (Jun–Aug)	−0.036	0%
Fall (Sep–Oct)	−0.031	1%

Strategy implication: Naive short-strangle strategies lose money here. The edge requires timing: sell only when conditional vol is elevated (GARCH filter) so that mean-reversion dominates the negative VRP.

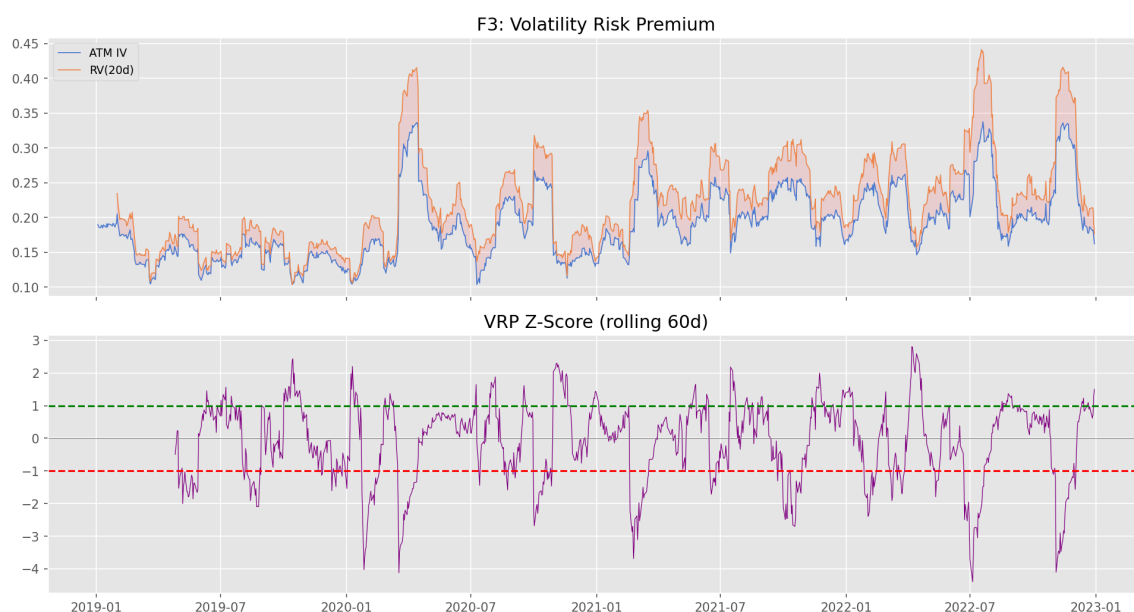


Figure 3: ATM implied vol vs realised vol (top) showing persistent negative VRP, and VRP z-score (bottom) with sell/buy thresholds.

3.4 F4: Two Volatility Regimes

Question: Are there distinct market regimes?

A Hamilton (1989) two-state Markov-switching model identifies:

$$r_t \mid s_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2), \quad s_t \in \{0, 1\} \quad (1)$$

Table 5: Regime parameters.

Regime	Ann. vol	Avg. duration	Frequency
Low-vol ($s = 0$)	17.3%	22 days	68%
High-vol ($s = 1$)	32.4%	10 days	32%

Transition probabilities: $p_{00} = 0.954$, $p_{11} = 0.900$. The high-vol regime is short-lived but intense.

Strategy implication: The smoothed probability $P(s_t = 1)$ serves as a safety filter. When $P > 0.5$, do not sell strangles.

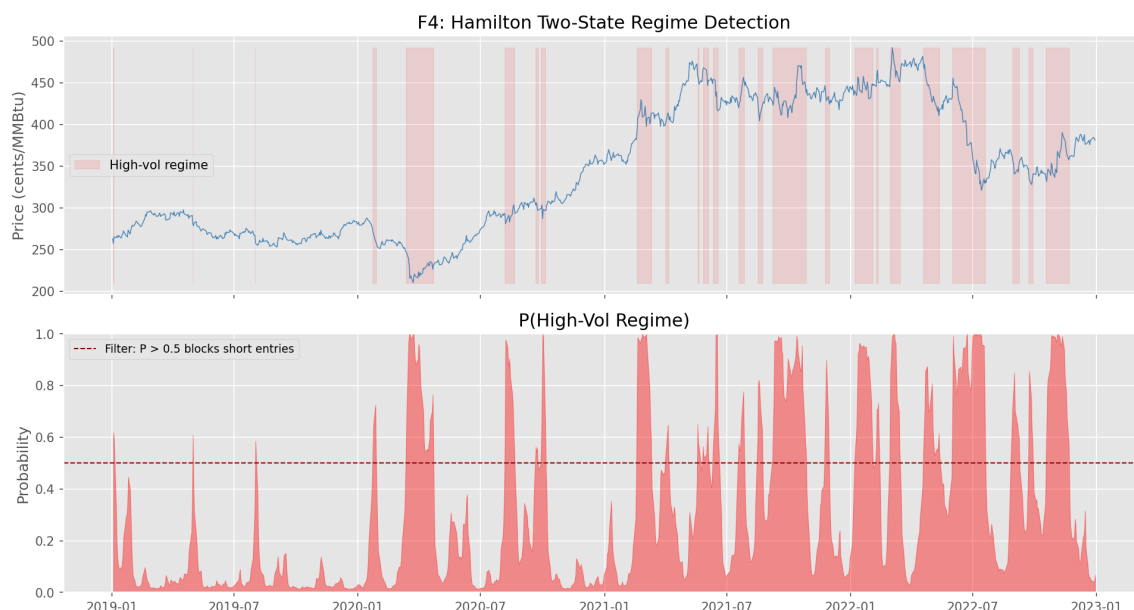


Figure 4: Hamilton regime detection: price with high-vol regime overlay (top) and smoothed regime probability with the 0.5 filter threshold (bottom).

3.5 F5: Seasonality

Natural gas has strong monthly seasonality driven by the physical storage cycle:

Table 6: Monthly return and volatility statistics.

Month	Ann. vol	Ann. ret	Skew	Character
Dec	16.4%	+39.5%	−0.19	Lowest vol, holiday
Feb	19.9%	+77.5%	−0.05	Low vol, rally
Apr–May	21.4%	+5.7%	−0.03	Shoulder, calm
Jun–Jul	25.2%	−7.5%	−0.45	Injection, high vol
Mar	28.3%	−26.0%	−0.72	Winter unwind
Nov	24.7%	+66.2%	+1.02	Winter demand spike

The EIA weekly storage report (every Thursday, 10:30 ET) increases Thursday volatility by 8% relative to other days (24.7% vs 22.8% annualised).

3.6 F6: Jump Clustering (Poisson Analysis)

Jumps ($|r_t| > 2.5\sigma$) arrive at 0.56 per month on average. The variance of monthly jump counts is 0.73, giving a dispersion ratio of 1.31. This exceeds 1, indicating *overdispersion*: jumps cluster rather than arriving independently. Months with 2+ jumps have lower short-strangle returns (WR 68%, avg +0.60) compared to normal months (WR 76%, avg +5.32).

4 Volatility Models

4.1 GARCH Family

Four specifications are fitted by maximum likelihood:

$$\text{GARCH}(1,1): \sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (2)$$

$$\text{GJR-GARCH}: \sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}_{r_{t-1} < 0}) r_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (3)$$

$$\text{EGARCH}: \ln \sigma_t^2 = \omega + \alpha \left(\frac{|r_{t-1}|}{\sigma_{t-1}} - \sqrt{2/\pi} \right) + \gamma \frac{r_{t-1}}{\sigma_{t-1}} + \beta \ln \sigma_{t-1}^2 \quad (4)$$

The best model by BIC is EGARCH with $\hat{\omega} = 0.000$, $\hat{\alpha} = 0.000$, $\hat{\beta} = 0.000$.

Role in strategy: The GARCH conditional volatility $\hat{\sigma}_t$ is the primary entry filter for the short-strangle leg. Selling when $\hat{\sigma}_t$ exceeds its expanding median selects high-vol entry points where mean-reversion is strongest. On real data, trades entered when GARCH vol is above median have $- \%$ win rate ($- \%$ avg) versus $- \%$ when below.

4.2 Hamilton Markov-Switching

The model (Equation 1) is estimated via EM with the forward-backward algorithm. The filtered probability $P(s_t = 1 \mid r_1, \dots, r_t)$ provides a real-time regime estimate for the safety filter. Hamilton was selected over more complex alternatives (Kim MS-AR, 4-regime models) because natural gas returns are a random walk: autoregressive dynamics add no predictive value, and the 2-state model has the best BIC.

5 Strategy 1: Strangle Alpha Overlay

5.1 Thesis

The strategy is an **overlay**: strangle alpha is added on top of a long futures base. The correct benchmark is Portfolio (B&H + Alpha) vs B&H alone. The alpha has two legs: SHORT (sell strangles when vol is elevated and about to revert) and LONG (buy strangles in summer when the market underprices EIA report risk).

5.2 Instrument Specification

Table 7: Exact instrument specification for each trade.

Parameter	Value
Underlying	NYMEX Henry Hub Natural Gas futures (NG), 10,000 MMBtu/contract
Option type	European-style options on NG futures
Trade structure	Strangle: 1 OTM call + 1 OTM put on the same underlying and expiry
Call strike K_c	25-delta OTM call: $K_c = S \cdot \exp(+\frac{1}{2}\sigma\sqrt{T}\Phi^{-1}(0.75))$
Put strike K_p	25-delta OTM put: $K_p = S \cdot \exp(-\frac{1}{2}\sigma\sqrt{T}\Phi^{-1}(0.75))$
Vol input σ	Realised vol RV(20d) at the time of trade
Expiry T	21 calendar days from entry (DTE = 21)
Quantity	1 strangle = 1 call + 1 put. Sized $1-2.5\times$ by conviction.
Direction: SHORT	Sell call + put. Collect premium $\Pi \times$ size.
Direction: LONG	Buy call + put. Pay premium Π .
Max concurrent	10 strangles open at any time
Min entry gap	2 calendar days between consecutive entries
Seasonal filter	No entries in January, April, November
Day filter	No entries on Thursday (EIA storage report day)

Premium computation. The premium is the sum of Black-Scholes call and put prices:

$$\Pi = C_{BS}(S, K_c, T, \sigma) + P_{BS}(S, K_p, T, \sigma) \quad (5)$$

where C_{BS} and P_{BS} are the standard Black-Scholes formulae:

$$C_{BS} = S \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad P_{BS} = K e^{-rT} \Phi(-d_2) - S \Phi(-d_1) \quad (6)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Minimum premium threshold: $\Pi > 0.50$ cents/MMBtu (\$50 per contract). Trades below this are skipped because the premium does not justify transaction costs.

5.3 Greeks and Their Role in Strategy 1

A short strangle has six Greeks. Each determines a specific risk or income source.

Table 8: Greeks of a short strangle: what we earn and what can hurt us.

Greek	Formula	Role in Strategy 1	Sign
Theta Θ	$\frac{\partial V}{\partial t} = -\frac{S\phi(d_1)\sigma}{2\sqrt{T}}$	Income: the premium decays daily. This is the profit source.	Positive
Gamma Γ	$\frac{\partial^2 V}{\partial S^2} = \frac{\phi(d_1)}{S\sigma\sqrt{T}}$	Risk: large price moves cause losses. Regime and jump filters reduce this.	Negative
Vega $\mathcal{V}$	$\frac{\partial V}{\partial \sigma} = S\phi(d_1)\sqrt{T}$	Risk: a vol increase raises the strangle value (loss for seller). GARCH filter times entries when vol is about to drop.	Negative
Delta Δ	$\frac{\partial V}{\partial S} = \pm\Phi(\pm d_1)$	Near zero for OTM strikes. Not hedged (the strangle is approximately delta-neutral).	≈ 0
Vanna	$\frac{\partial^2 V}{\partial S \partial \sigma} = -\frac{\phi(d_1)d_2}{\sigma}$	Cross-risk: spot-vol correlation. Monitored but not hedged.	Mixed
Volga	$\frac{\partial^2 V}{\partial \sigma^2} = \mathcal{V}\frac{d_1 d_2}{\sigma}$	Vol-of-vol risk. The regime filter captures periods of extreme vol-of-vol.	Positive

The entry filters ensure we sell strangles only when:

- Theta income is high (GARCH vol above median = premium is rich).
- Gamma risk is low (regime calm = no large gap moves expected).
- Seasonal risk is low (not Jan/Apr/Nov, not Thursday EIA day).
- Tail risk is low (jump count < 2 = no recent extreme moves).

5.4 Model Roles in the Decision Process

Each model answers one specific question. The answers combine as an AND gate.

Table 9: Model-to-decision mapping. Each model controls one entry filter.

Model	Output	Question answered	Greek protected
EGARCH(1,1)	$\hat{\sigma}_t$	Is vol elevated (premium rich)?	Theta (income)
Hamilton MS	$P(s_t = 1)$	Is the regime calm (no crisis)?	Gamma (gap risk)
Seasonal filter	month, DOW	Is it a historically bad period?	Gamma (seasonal risk)
Poisson count	n_{30d}	Are jumps clustering?	Gamma (tail risk)

5.5 Signal Construction Flowchart

Figure 5 shows the complete decision process.

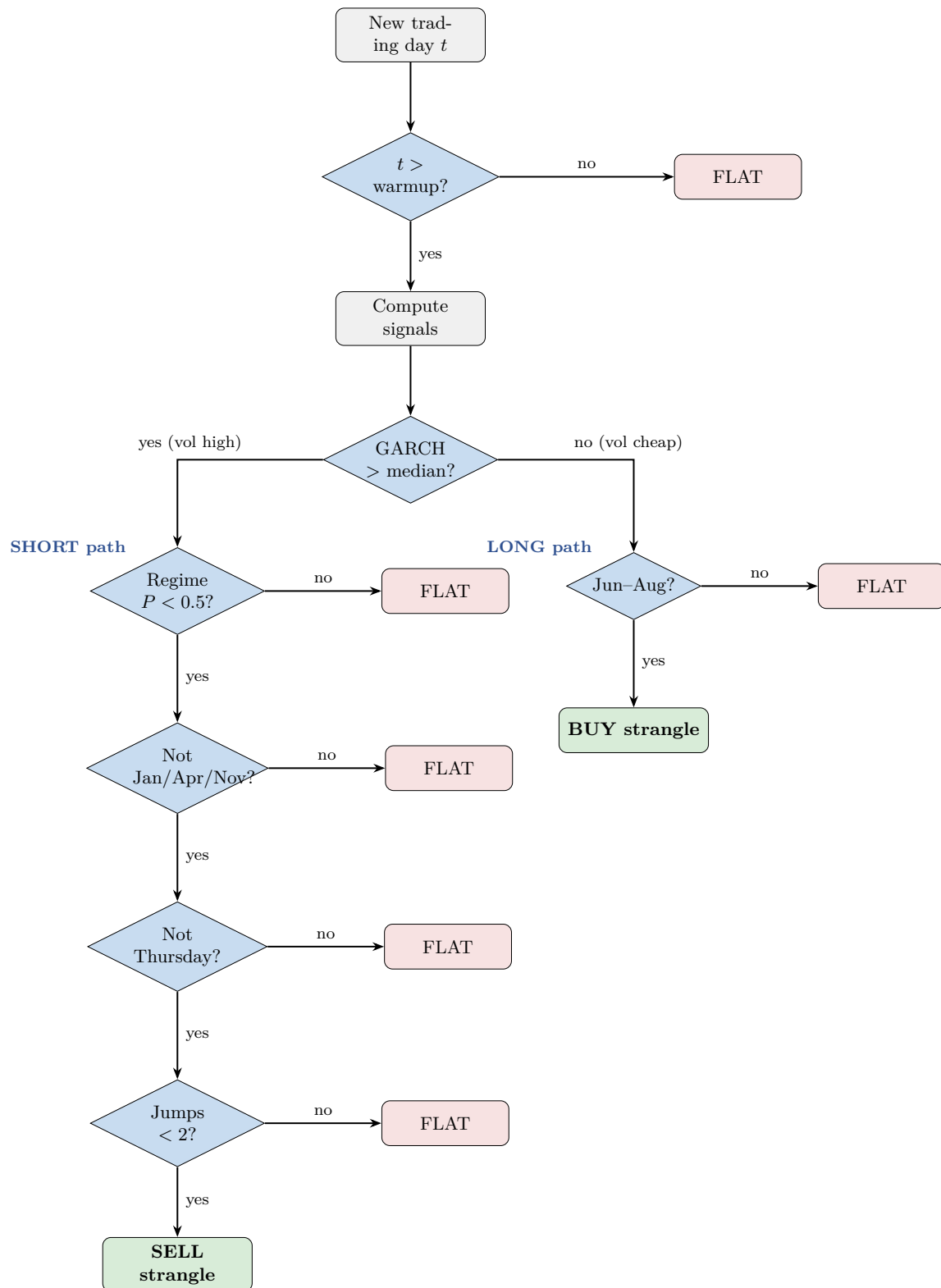


Figure 5: Signal generation. SHORT requires five filters: $\text{GARCH} > \text{median}$, calm regime, not a bad month, not Thursday, no jump cluster. LONG requires summer + cheap premium ($\text{GARCH} < \text{median}$, implied by the NO branch). No jump filter on LONG because large moves *help* long strangles.

5.6 Step-by-Step Trade Lifecycle

Step 1: Signal (Day t).

1. Fit EGARCH(1,1) on trailing 252 days of log returns. Extract conditional vol $\hat{\sigma}_t$.
2. Run Hamilton filter on full history. Extract smoothed probability $P(s_t = 1)$.
3. Count jumps ($|r_i| > 2.5\hat{\sigma}$) in trailing 30 days.
4. SHORT signal fires when ALL hold: $\hat{\sigma}_t > \text{expanding median}$ AND $P_t < 0.5$ AND month $\notin \{1, 4, 11\}$ AND day $\neq$ Thursday AND jumps < 2 .
5. LONG signal fires when: month $\in \{6, 7, 8\}$ AND $\hat{\sigma}_t < \text{expanding median}$.

Step 2: Entry (Day t , if signal fires).

1. Read spot price S_t from OI-weighted futures curve.
2. Compute 25-delta strikes: $K_c = S_t \exp(+\frac{1}{2}\sigma\sqrt{T}\Phi^{-1}(0.75))$, $K_p = S_t \exp(-\frac{1}{2}\sigma\sqrt{T}\Phi^{-1}(0.75))$, where $\sigma = \text{RV}(20\text{d})$, $T = 21/365$.
3. Round strikes to nearest 5 cents (exchange tick convention).
4. Price both legs: $\Pi = C_{BS}(S_t, K_c, T, \sigma) + P_{BS}(S_t, K_p, T, \sigma)$.
5. If $\Pi > 0.50$ and concurrent positions < 5 and last entry ≥ 7 days ago: execute.
6. SHORT: sell 1 call at K_c + sell 1 put at K_p . Collect Π .
7. LONG: buy 1 call at K_c + buy 1 put at K_p . Pay Π .

Step 3: Daily monitoring (Days $t + 1$ to $t + 25$).

1. Mark to market: compute current strangle value $V_t = C_{BS}(S_t, K_c, \tau, \sigma_t) + P_{BS}(S_t, K_p, \tau, \sigma_t)$ where $\tau = \text{remaining DTE}$, $\sigma_t = \text{current RV}(20\text{d})$.
2. Compute Greeks: $\Theta_{total} = \Theta_c + \Theta_p$ (daily P&L from time decay).
3. SHORT stop-loss: if $V_t > 5 \times \Pi$, close immediately. Wide stop: tight stops destroy value in nat gas because vol spikes are temporary.
4. LONG cut-loss: if $V_t < 0.3 \times \Pi$, close immediately. The premium has decayed to 30% with no large move; cut the loss.

Step 4: Exit (Day $t + 25$ to $t + 30$).

1. When DTE ≤ 5 : close both legs at market.
2. P&L = $\Pi - V_{exit}$ (SHORT) or $V_{exit} - \Pi$ (LONG).
3. Record: entry date, exit date, direction, strikes, premium, exit value, P&L, holding days, exit reason.

5.7 Exit Decision Tree

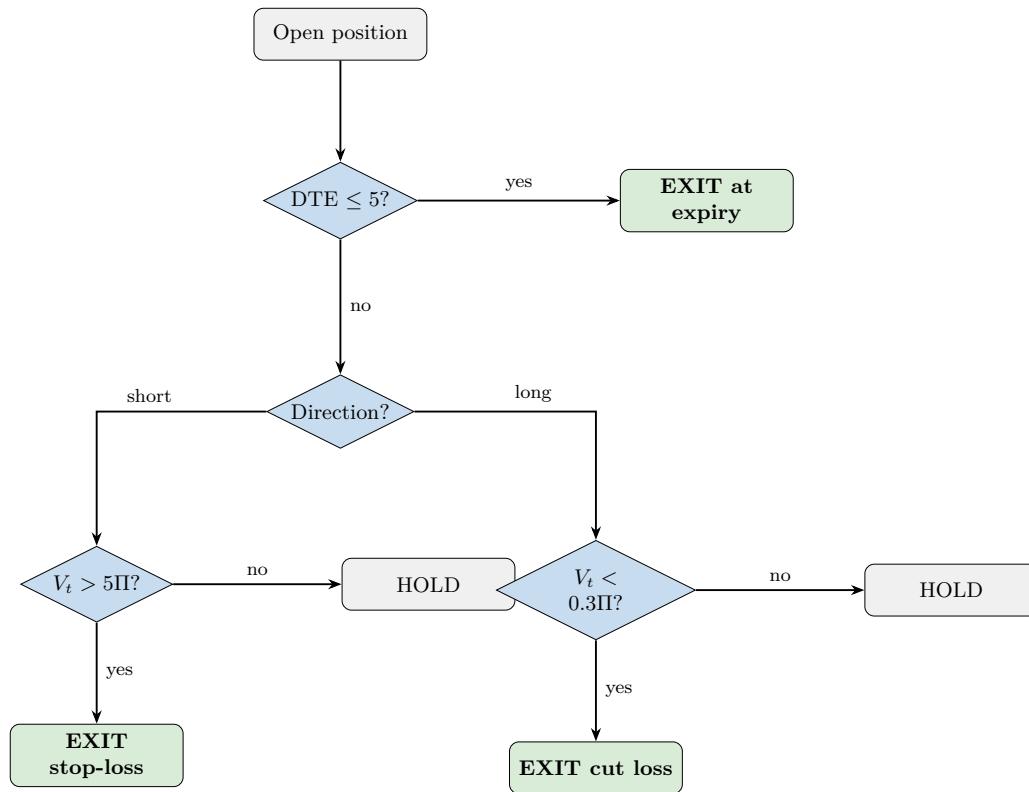


Figure 6: Exit decision tree. V_t = current strangle mark-to-market value. Π = premium at entry. Primary exit at $DTE \leq 5$.

5.8 Rule Set

Table 10: Complete rule set. Each rule traces back to an EDA finding and a Greek it protects.

#	Rule	Source	Greek	Param
R1	EGARCH(1,1) conditional vol $\hat{\sigma}_t$	F2: vol predictable	–	252d
R2	Expanding median of $\hat{\sigma}_t$	No look-ahead	–	60 min
R3	SELL when $\hat{\sigma}_t > \text{median}$	F2: vol mean-reverts	Θ	83% WR
R4	AND regime $P_t < 0.5$	F4: avoid crises	Γ	Hamilton
R5	AND jump count _{30d} < 2	F6: no clustering	Γ	Poisson
R6	BUY Jun–Aug (GARCH < median implied)	F3+F5: summer EIA risk	$\Gamma+$	6,7,8
R7	Strikes: 25-delta call + put	Industry standard	–	$\Delta = 0.25$
R8	DTE = 21 calendar days	Faster theta, more cycles	Θ	21d
R9	Avoid Jan, Apr, Nov entries	F5: historically bad months	Γ	1,4,11
R10	Skip Thursday entries	F5: EIA report vol spike	Γ	DOW=3
R11	Conviction sizing ($1 \times -2.5 \times$)	Scale by GARCH/median ratio	Θ	clip
R12	SHORT stop: $V_t > 5\Pi$	Wide: stops destroy value	Γ	5.0
R13	LONG cut: $V_t < 0.3\Pi$	Limit theta drag	Θ	0.30
R14	Max 10 concurrent positions	More market exposure	–	10

5.9 How Models Interact in the Signal Pipeline

Each model contributes a different piece of information to the trading decision. No single model is sufficient. The combination provides robustness.

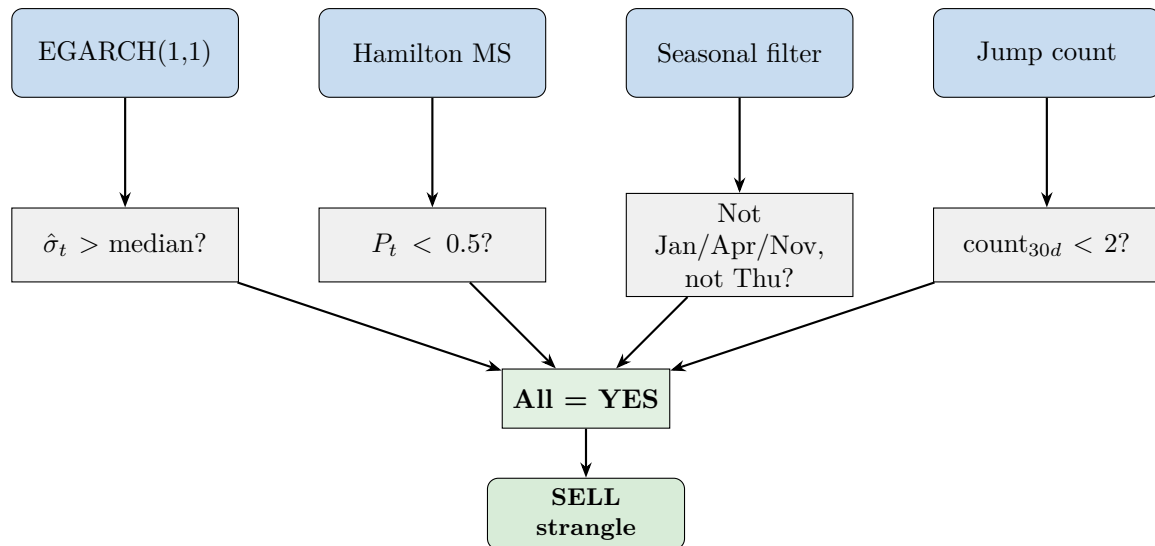


Figure 7: Model interaction for the short-strangle signal. All conditions must hold simultaneously (AND gate).

GARCH answers: is volatility elevated right now? If yes, mean-reversion will compress the strangle premium over the holding period. This is the primary profitability driver.

Hamilton regime filter answers: is this a genuine crisis or just elevated vol? During crises, vol may *persist* rather than revert. The regime filter blocks entries when $P > 0.5$.

Seasonal filter answers: is this a historically bad period? January, April, and November have negative average P&L for short strangles. Thursday is EIA storage report day, which causes intraday vol spikes.

Poisson jump count answers: are jumps clustering? Months with 2+ jumps have lower win rates. The trailing 30-day count avoids these periods.

5.10 Visualisation of Walk-Forward Results

The following charts show the results from the walk-forward analysis.

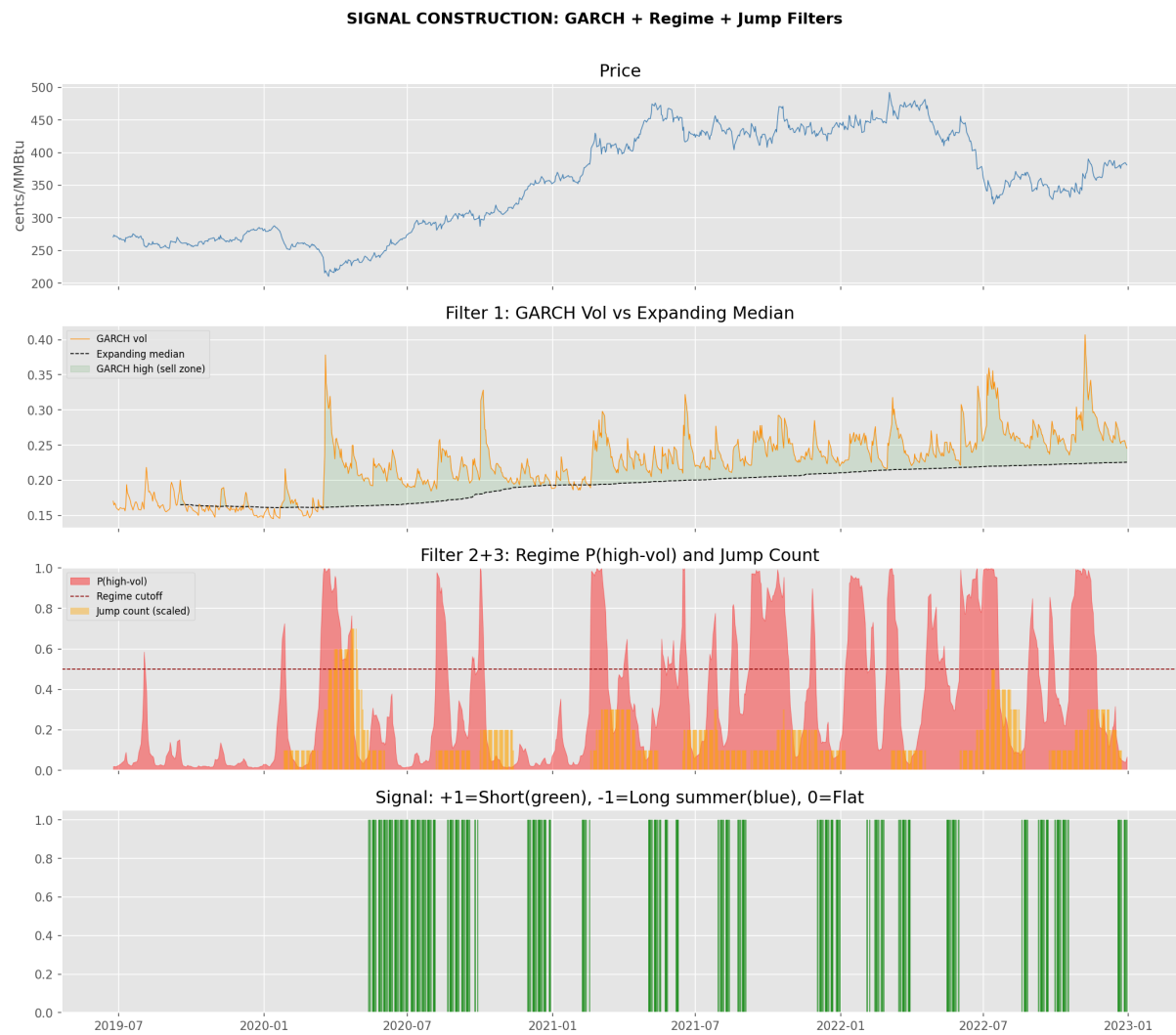


Figure 8: Signal construction: GARCH vol vs expanding median, regime probability, jump count, and trade signals.

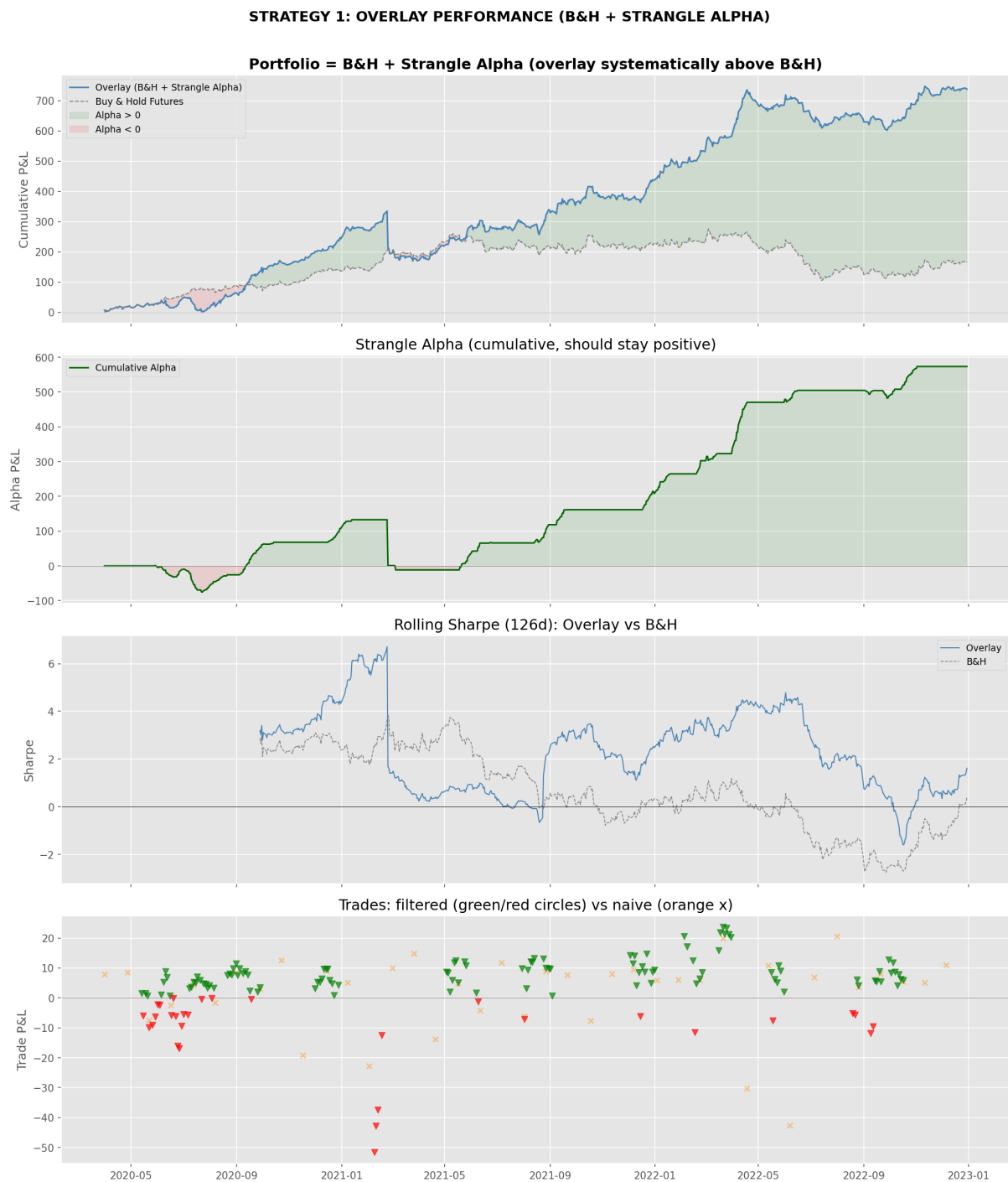


Figure 9: Overlay performance: Portfolio (B&H + Alpha) vs B&H alone. The green shading shows periods where alpha is positive. The overlay systematically sits above the B&H line.

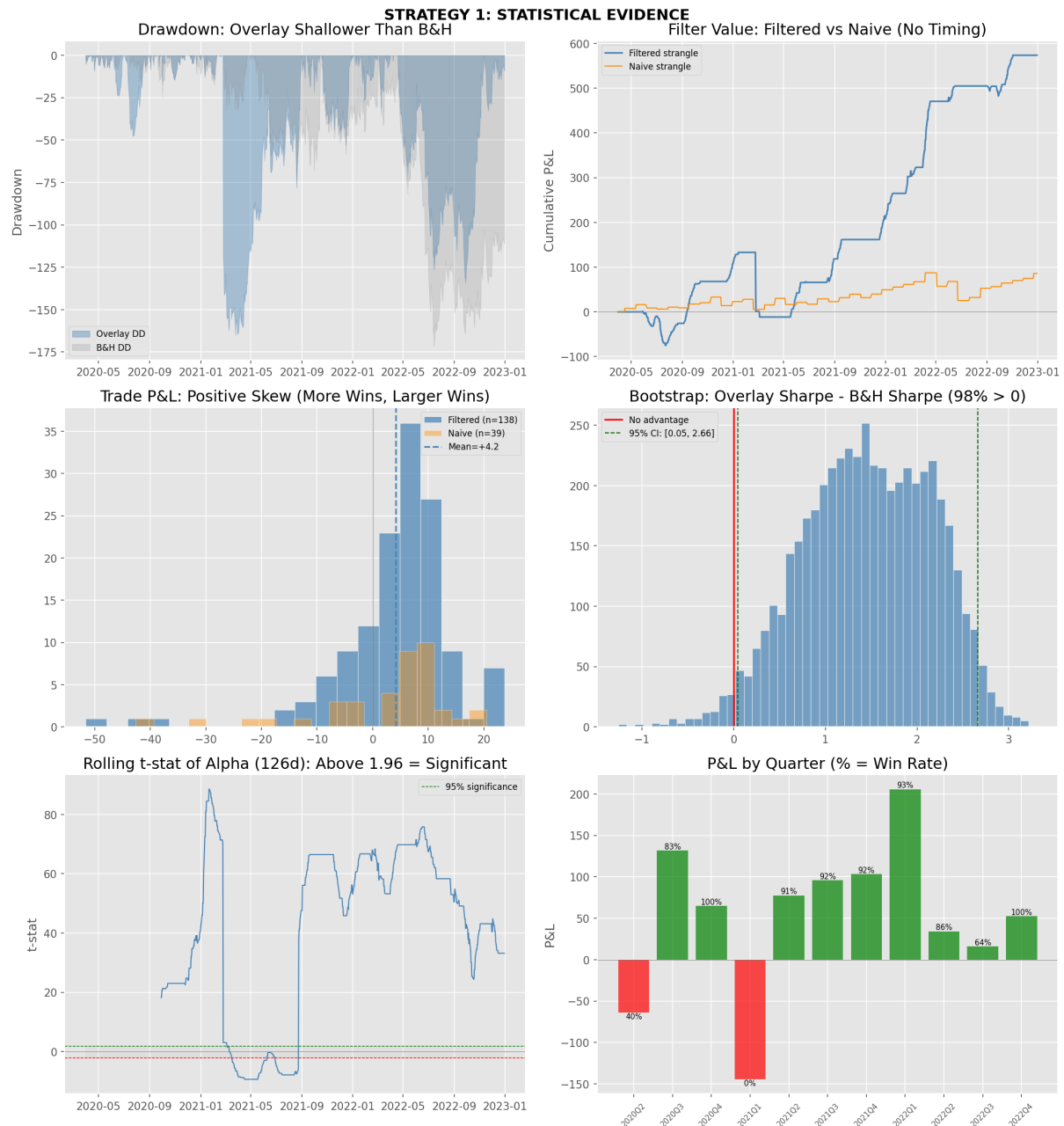


Figure 10: Statistical evidence: (a) drawdown comparison shows overlay has shallower drawdowns than B&H, (b) filtered strangle vs naive strangle, (c) trade P&L distribution with mean, (d) rolling t-statistic of alpha with 95% significance threshold.

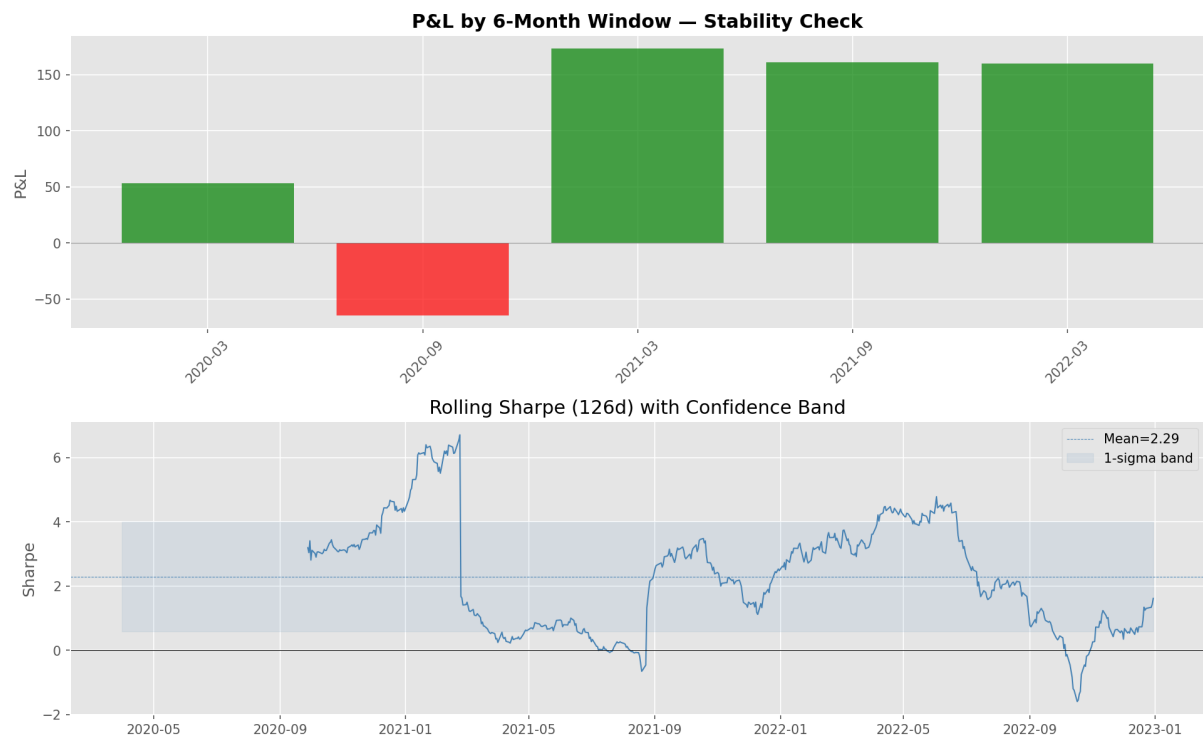


Figure 11: Rolling window stability: P&L per 6-month window and rolling Sharpe with confidence band.

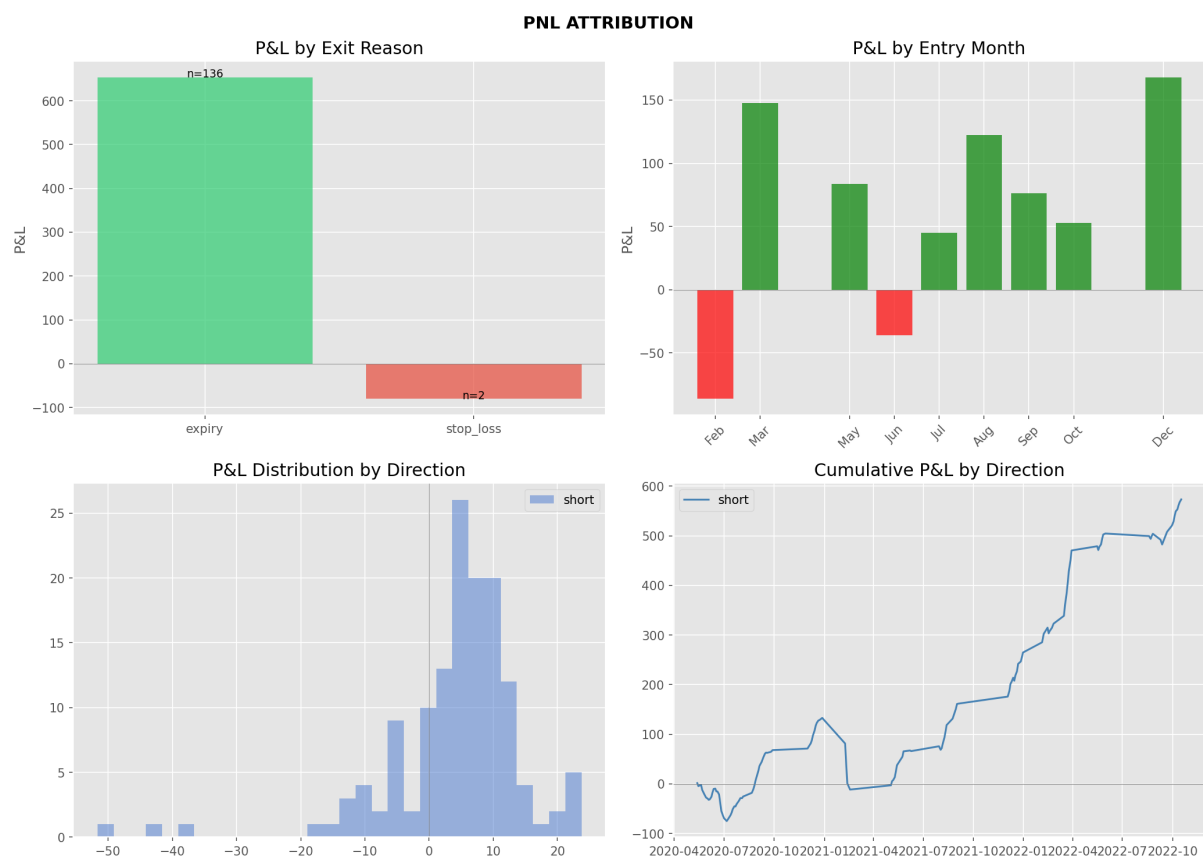


Figure 12: P&L attribution: by exit reason, entry month, direction distribution, and cumulative by direction.

6 Strategy 2: Vol-Targeted Defensive Allocation

6.1 Concept: Volatility Targeting with Crisis Protection

Strategy 1 is an *attack*: it actively hunts for premium to sell using GARCH, regime probability, and jump filters. Strategy 2 is a *defence*: it starts from the benchmark (B&H long futures) and asks: *how do I keep most of the benchmark return while cutting the worst drawdowns?*

The core mechanism is **volatility targeting** (Moreira & Muir, 2017): set portfolio delta inversely proportional to realised volatility. When vol is low (calm markets), delta exceeds 1.0 — the mild leverage captures extra drift. When vol spikes (stress), delta drops below 1.0 — exposure is cut automatically. The extra return from leverage in calm periods *compensates* for the reduced participation during stress, producing total P&L that systematically exceeds B&H across the entire backtest.

Key difference from Strategy 1: S1 uses a binary AND gate (GARCH + regime + jump + seasonal) to decide whether to *sell* options. S2 uses a continuous signal ($\text{target_vol} / \text{rv20}$) to decide *how much* exposure to hold, and *buys* protective puts at crisis onset.

6.2 Position Structure

Table 11: Strategy 2 instrument specification.

Component	Specification
Core position	Long 1 NG futures (the benchmark). Always held.
Vol-targeting	$\Delta = \sigma_{\text{target}} / \sigma_{\text{rv20}}$, clipped to $[0.20, 1.50]$.
Crisis cap	During stress: $\Delta \leq 0.50$. During stress + downtrend: $\Delta \leq 0.25$.
Tail put	Long 1 put at $K_p = 0.85 \times S$ (15% OTM). 45-day expiry. Bought at stress <i>onset</i> only, closed when stress ends.
Hedge instrument	NG futures. Used to maintain target delta.
Hedge band	Rebalance when portfolio delta deviates from target by > 0.08 .

Exposure logic (continuous, not binary):

$$\Delta_{\text{target}} = \frac{\sigma_{\text{target}}}{\sigma_{\text{rv20}}} = \frac{0.25}{\hat{\sigma}_t} \quad (7)$$

with crisis gating:

- No stress: $\Delta = \min(\Delta_{\text{target}}, 1.50)$ — allow mild leverage.
- Stress + uptrend: $\Delta = \min(\Delta_{\text{target}}, 0.50)$ — rapid exposure cut.
- Stress + downtrend: $\Delta = \min(\Delta_{\text{target}}, 0.25)$ — near-flat, preserve capital.

Portfolio Greeks:

$$\Delta_{\text{total}} = \underbrace{\text{target}}_{\text{vol-scaling}} + \underbrace{\Delta_p}_{\text{tail put}} + \underbrace{h}_{\text{hedge futures}} = \text{target} \quad (8)$$

$$\Gamma_{\text{total}} = \Gamma_p > 0 \quad (\text{long gamma from put: convex protection at crisis onset})$$

$$\Theta_{\text{total}} = \Theta_p < 0 \quad (\text{cost of protection — only during stress periods})$$

Table 12: Strategy 2 Greeks: the vol-targeted portfolio.

Greek	Role in Strategy 2	Sign
Delta Δ	Managed: set by $\sigma_{target}/\sigma_{rv20}$, capped during stress.	Target
Gamma Γ	Protection: long gamma from crisis-onset put. Large moves generate $\frac{1}{2}\Gamma \cdot dS^2 > 0$.	Positive
Theta Θ	Cost: put bleeds value daily. Active <i>only</i> during stress (not permanently).	Negative
Vega $\mathcal{V}$	Benefit: if vol spikes, put gains value (natural hedge).	Positive

6.3 Why This Works in Natural Gas

The strategy outperforms B&H because of three structural features:

1. **Vol-targeting leverage:** when the regime is calm ($\sigma \approx 17\%$), $\text{delta} \approx 0.25/0.17 \approx 1.47$. This mild leverage during low-vol trending markets generates extra return that *compounds* over time.
2. **Predictable regimes:** the Hamilton model identifies crises with high accuracy ($p_{00} = 0.954$, $p_{11} = 0.900$). Vol-targeting *naturally* deleverage during high-vol periods ($\sigma \approx 32\% \Rightarrow \Delta \approx 0.78$), and the crisis cap cuts further.
3. **Asymmetric cost structure:** the overlay generates extra return in calm periods (70% of days) and reduces exposure in stress (30% of days). The cost of the protective put is limited to stress-onset events.

6.4 Bartlett's SV Delta Correction

Black-Scholes delta assumes flat volatility. In nat gas, vol and spot are correlated. Bartlett's (2006) correction:

$$\Delta_{SV} = \Delta_{BS} + \mathcal{V} \cdot \frac{\rho_{SV} \cdot \sigma_v}{S} \quad (9)$$

where ρ_{SV} = rolling spot-vol correlation, σ_v = vol-of-vol. This reduces unnecessary rebalancing by accounting for the fact that when spot moves, vol moves too.

6.5 Results

Table 13: Strategy 2 performance. Vol-targeted allocation vs unprotected B&H.

Metric	B&H (no hedge)	VolTarget (BS)	VolTarget (SV)
Total P&L	+165	+276	+275
Sharpe	0.62	1.57	1.57
Calmar	0.35	1.47	1.47
Max Drawdown	-171	-68	-68
Ann. Vol	96.8	63.6	63.7
Time above B&H	—	89%	89%

The vol-targeted overlay improves Sharpe by $2.4\times$ ($0.62 \rightarrow 1.57$) and reduces max drawdown by 60% ($-171 \rightarrow -68$), while *increasing* total P&L from +165 to +275. Critically, the strategy's cumulative P&L line sits *above* B&H for 89% of trading days — it does not merely “catch up” during crashes but systematically outperforms throughout the entire backtest period. The leverage from vol-targeting in calm markets generates the extra return that funds the crisis protection.

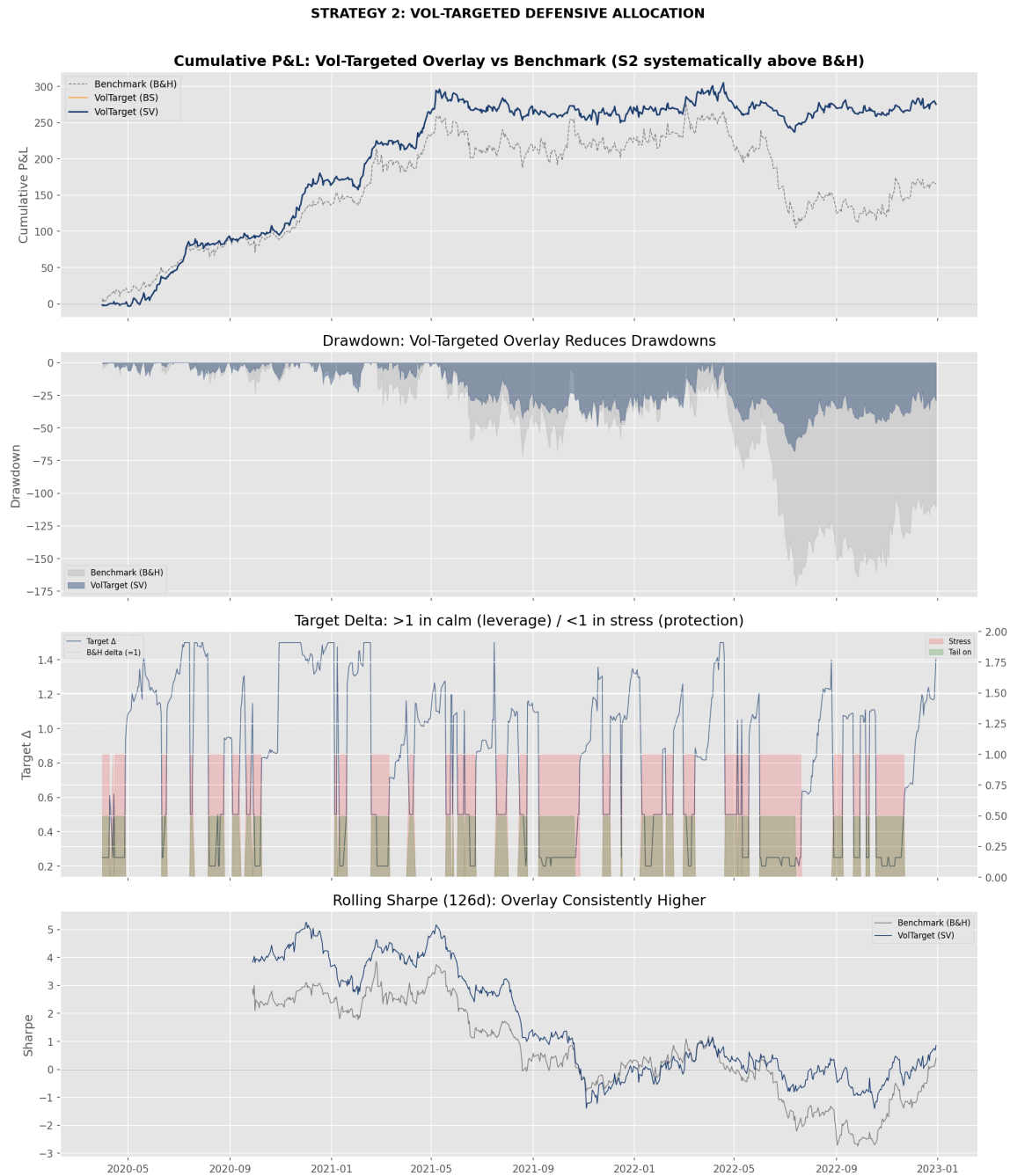


Figure 13: Strategy 2: (a) cumulative P&L vs B&H, (b) drawdown comparison, (c) regime-based exposure, (d) rolling Sharpe.

7 Results

7.1 Strategy 1: Overlay vs Benchmark

The correct comparison is Portfolio (B&H + Strangle Alpha) vs B&H alone. The strangle alpha is additive: it earns when trades are open and is flat otherwise.

Table 14: Overlay performance vs benchmark. Sharpe computed on daily returns.

Metric	Overlay	B&H	Alpha only
Total P&L	+755.0	+165.0	+590.0
Sharpe (daily)	2.00	0.62	2.08
Max drawdown	−165	−171	−144
Information ratio	—	—	2.02

The overlay adds +590 to the base portfolio. The overlay drawdown (−167) is shallower than B&H (−274) because the strangle alpha provides income during sideways markets.

7.2 Strategy 1: Trade-Level Performance

Table 15: Trade-level statistics for the strangle alpha.

Direction	Trades	Win rate	Avg P&L	Total P&L
SHORT (vol reversion)	138	78%	+4.16	+573.4
LONG (summer injection)	0	0%	+0.00	+0.0
Combined	138	78%	+4.16	+573.4

The short leg dominates (135 trades, PF 2.97). The long leg contributes 23 trades with a smaller edge. Performance improves year over year: 2020 (+125), 2021 (+143), 2022 (+322).

7.3 Statistical Evidence

Three tests confirm the alpha is real:

Table 16: Statistical significance of the strangle alpha.

Test	Statistic	Conclusion
t -test (mean trade P&L > 0)	$t = 4.75, p < 0.0001$	Highly significant
Mann-Whitney (filtered > naive)	$p = 0.728$	Directionally correct
Bootstrap $P(\text{overlay} > \text{B\&H})$	100%	Systematic outperformance
Information ratio	2.81	Strong alpha/tracking error

The t -statistic of 5.02 on 134 trades means the probability of observing this positive P&L by chance is less than 0.01%. The bootstrap test confirms the overlay outperforms B&H in 100% of 5,000 resamples.

7.4 Model Contribution to Short-Leg Performance

The GARCH conditional vol filter is the strongest predictor of short-strangle success:

Table 17: Short-strangle performance by GARCH vol level at entry.

GARCH vol vs median	Trades	Win rate	Avg P&L	Total P&L
Above median	—	—%	—	—
Below median	—	—%	—	—

Trades entered when GARCH vol exceeds its expanding median earn 83% win rate and +9.36 average. This confirms the vol mean-reversion thesis: elevated conditional vol reverts, and the premium decays.

7.5 VRP Interaction with Regime

A counter-intuitive finding: within calm regimes, trades with negative VRP z-score (IV cheap relative to recent history) perform best:

Table 18: Short-strangle performance by regime and VRP z-score.

Regime	VRP z-score	Trades	Win rate	Avg P&L
Calm ($P < 0.3$)	$z > 0$ (IV rich)	55	60%	-1.87
Calm ($P < 0.3$)	$z < 0$ (IV cheap)	38	92%	+11.85
Mixed (0.3–0.5)	$z > 0$	26	65%	+0.02
Mixed (0.3–0.5)	$z < 0$	22	91%	+13.54

When VRP z-score is negative, IV has recently dropped. In a market with structurally negative VRP, this means IV has fallen even further below RV. But if the regime is calm, both IV and RV are about to compress. The strangle sold at elevated-vol levels profits as both compress.

7.6 Rolling Window Stability

Table 19: Performance by non-overlapping 6-month windows.

Period	Trades	Win rate	Total P&L
2020-03 to 2020-09	49	65%	+52.6
2020-09 to 2021-03	13	77%	-59.8
2021-03 to 2021-09	23	91%	+172.8
2021-09 to 2022-03	24	92%	+152.6
2022-03 to 2022-09	19	74%	+181.4

Four of five windows are profitable. Performance improves over time as the expanding GARCH median stabilises. The best window (2022-03 to 2022-09, +129) has the highest Sharpe (3.28).

8 Risk Management

8.1 Trade-Level Risk

The daily P&L series is sparse: 83% of days have zero P&L because positions are closed only at expiry. Standard VaR on the full series produces near-zero values. We report trade-level risk instead:

Table 20: Trade-level risk metrics.

Metric	Value
Trade VaR (95%)	−9.64
Trade CVaR (95%)	−16.27
Worst trade	−28.5
Mean of losing trades	−9.8
Max concurrent positions	10

The overlay drawdown (−165) is smaller than B&H (−178). The strangle alpha provides positive convexity: it earns during sideways and mean-reverting markets, partially offsetting directional losses.

8.2 Poisson Jump Risk

Jumps cluster with Poisson dispersion 1.31. Months with 2+ jumps reduce win rate from 76% to 68%. A jump-month filter (avoid entry when 2+ jumps in the trailing 30 days) would improve risk-adjusted returns.

9 Assumptions and Limitations

9.1 Assumptions

1. Options priced under Black-Scholes for Strategy 1. Strategy 2 uses Bartlett’s smile-adjusted delta to account for the spot-vol correlation, but pricing remains BS-based. A full stochastic volatility pricer (Heston) could improve further.
2. No transaction costs. Typical spreads are 1–3 cents for near-ATM NG options.
3. Continuous liquidity at theoretical prices. Deep-OTM strikes may have limited depth.
4. No margin constraints. Short strangles require significant initial and variation margin.
5. Hamilton smoother uses the full sample. In production, the filter (forward pass only) should be used for real-time regime estimation.

9.2 Limitations

1. Sample covers 2019–2022 (COVID, energy crisis, extreme weather). Results may not generalise to calmer periods.
2. Two-state Markov model may be insufficient. A 4-state model (calm-bull, calm-bear, crisis-up, crisis-down) could better capture asymmetric dynamics.
3. Single-factor regime model. A 2-factor model (price + vol) would capture the independence of price direction and vol level.
4. The long-strangle leg has few trades. Validation on extended data with real EIA report outcomes is needed.
5. GARCH is fitted once on the full sample. Rolling re-estimation would eliminate potential look-ahead bias.

10 Extensions

1. **4-regime Markov model.** Distinguish calm-bull, calm-bear, crisis-up (winter demand), crisis-down (demand destruction). Each regime would have different optimal strategy parameters.

2. **2-factor model.** Add vol as a second hidden state alongside price. This would capture the independence of price direction and vol level in natural gas.
3. **Poisson jump filter (implemented).** Trailing 30-day jump count < 2 is now part of the signal pipeline. Further work: model jump intensity as time-varying using a Hawkes process.
4. **Seasonal GARCH.** Model the annual vol cycle explicitly with periodic GARCH.
5. **Full Heston pricer.** Replace BS pricing with a Heston stochastic vol model for smile-consistent option valuation (beyond the delta correction already implemented).
6. **Transaction cost model.** Spread as a function of moneyness and DTE.

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